

“Local Update” issue for Pauli String sampling

We want to compute $S_{\text{SRE}} = -\log \Omega_N = -\log \left(\sum_P \frac{1}{2^N} \langle \psi | P | \psi \rangle^4 \right)$ where $\rho = |\psi\rangle\langle\psi|$

MCMC scheme

Let

$$\omega_P = \langle \psi | P | \psi \rangle^2$$

We can prove by the property of Pauli Group:

$$\sum_P \omega_P = 2^N$$

Thus we have

$$\Omega_N = \sum_P \left(\frac{\omega_P}{2^N} \right) \omega_P = \frac{1}{M} \sum_{i=1}^M \omega_{P_i}$$

where P_i are sampled from the distribution $\pi(P) = \frac{\omega_P}{2^N}$.

The update scheme is

$$\text{acc} = \min \left(1, \frac{\omega_{P_{\text{new}}}}{\omega_{P_{\text{old}}}} \right)$$

“Local Update” issue

How to sample through Pauli strings e.g.

$$P = X_1 Y_2 I_3 Z_4 \dots$$

We Propose

$$P' = Z_1 Y_2 I_3 Z_4 \dots = -i Y_1 P$$

which locally update one pauli character

thus the update ratio is

$$\frac{\omega_{P_{\text{new}}}}{\omega_{P_{\text{old}}}} = \frac{\langle \psi | P' | \psi \rangle^2}{\langle \psi | P | \psi \rangle^2}$$

Then the problem is, what if

$$P|\psi\rangle \text{ is a Stabilizer State of } iY \text{ Operator?}$$